Programming & Debugging Custom C2000 MCU Device Using On-board Debugger of a Texas Launchpad (Launchxl-F280049C)

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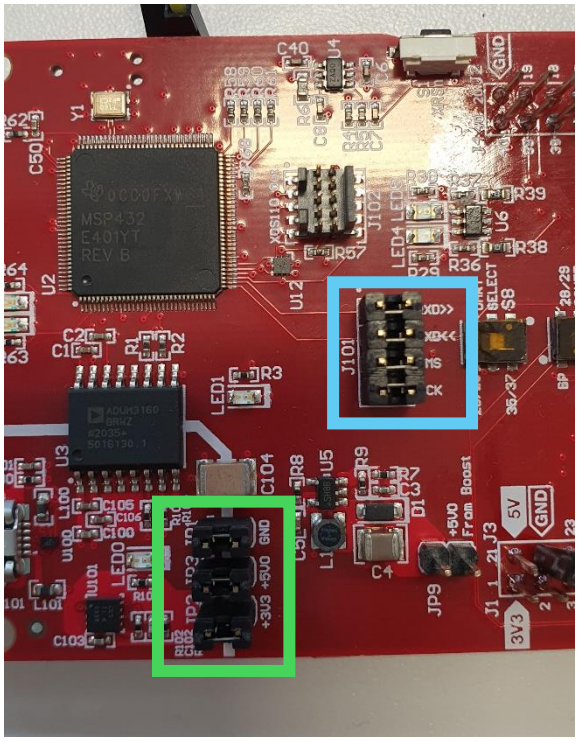
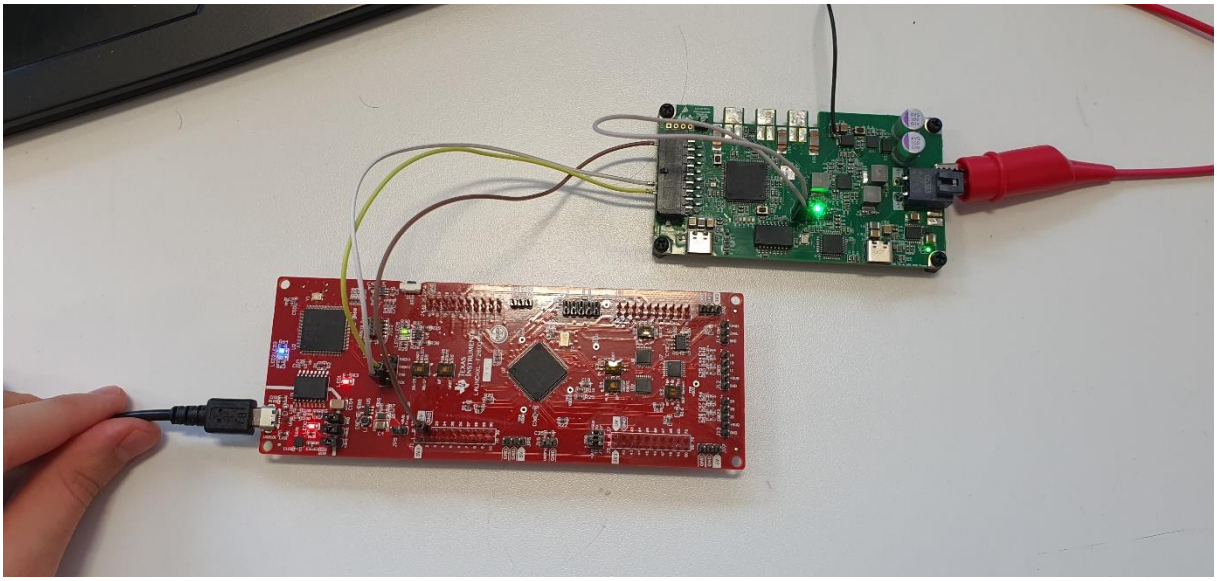
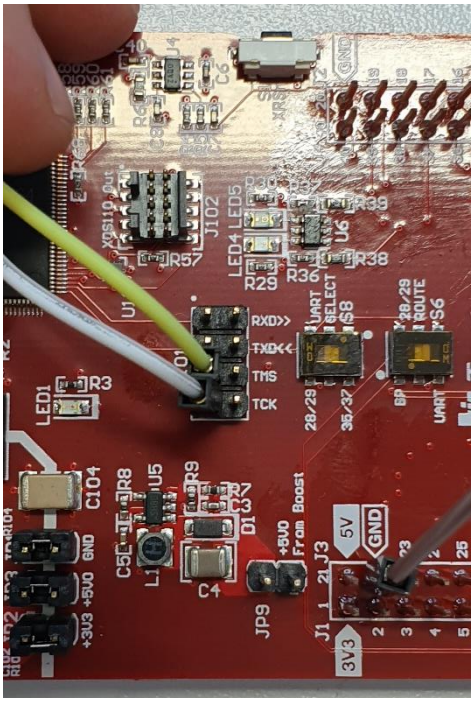
1) PREPARATION OF HARDWARE:

4 jumpers seen should be removed from the launchpad. (Blue rectangle) Also notice that 3 jumpers at the usb side should remain. (green rectangle)



TMS, TCK and GND connections should be made between the launchpad and the custom board.



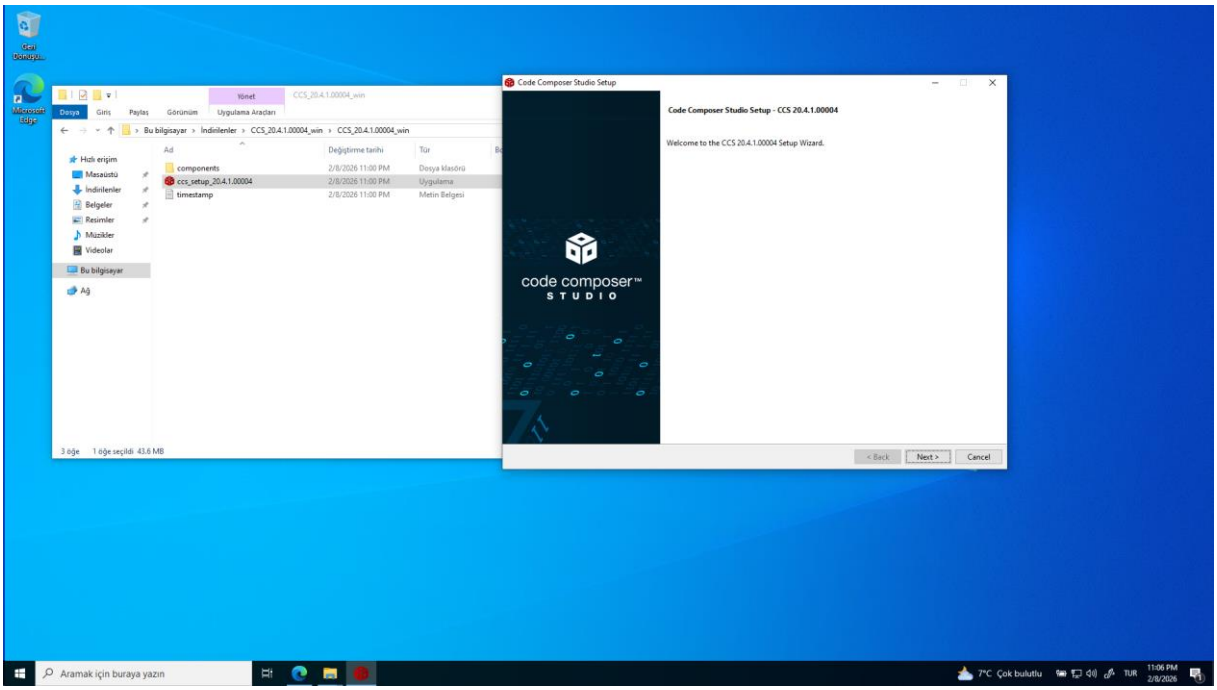


Then supply power to your custom board and connect the usb connection of launchpad to pc.

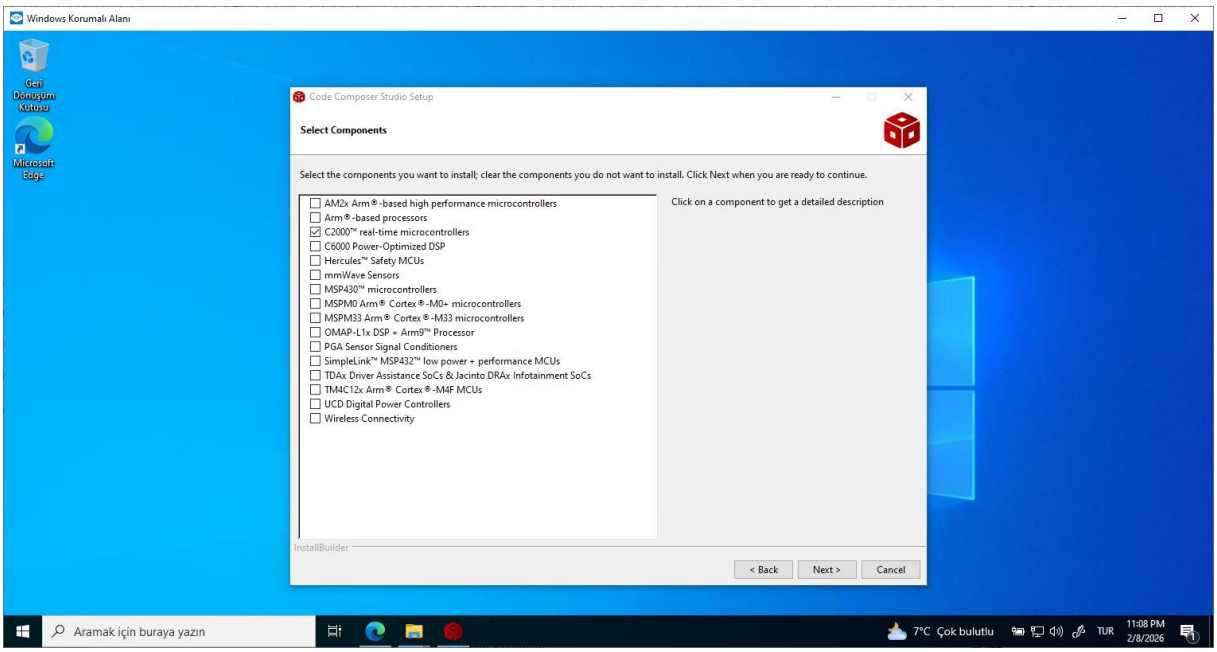
2) PREPARATION OF SOFTWARE:

From the https://www.ti.com/tool/CCSTUDIO download Code Composer Studio (CCS) IDE.

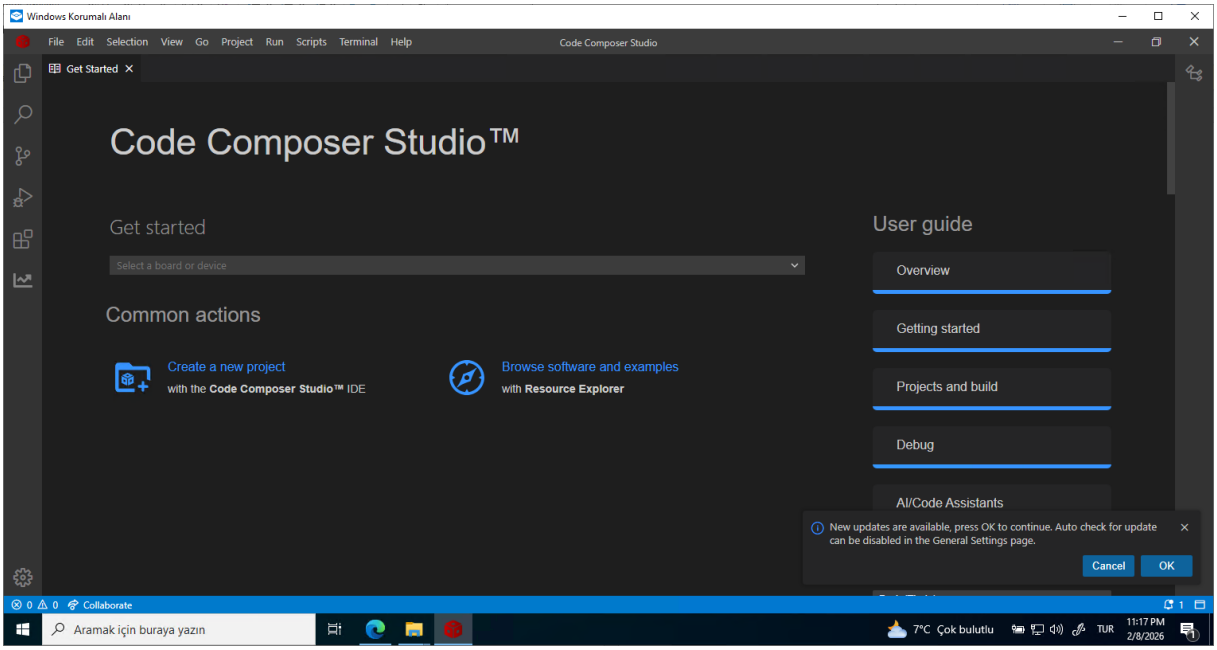
Run the ccs_setup.exe and install the app by following the setup procedure.



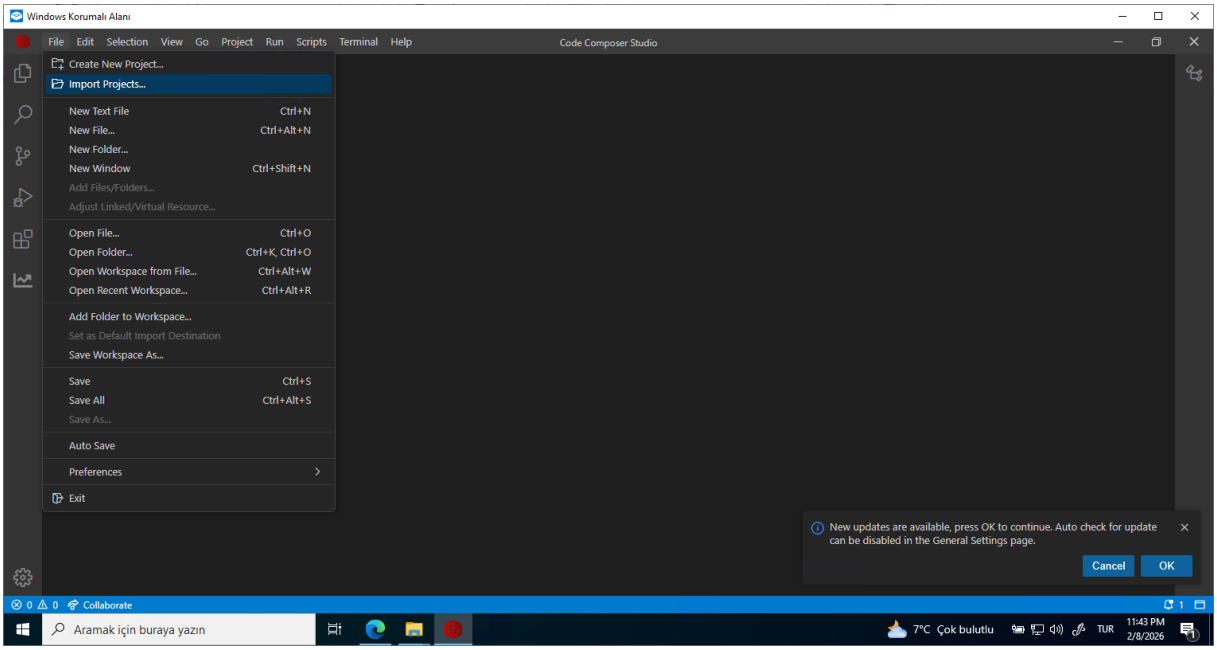
At the select component window select only the C2000 real-time microcontrollers and continue.



Then run CCS IDE.

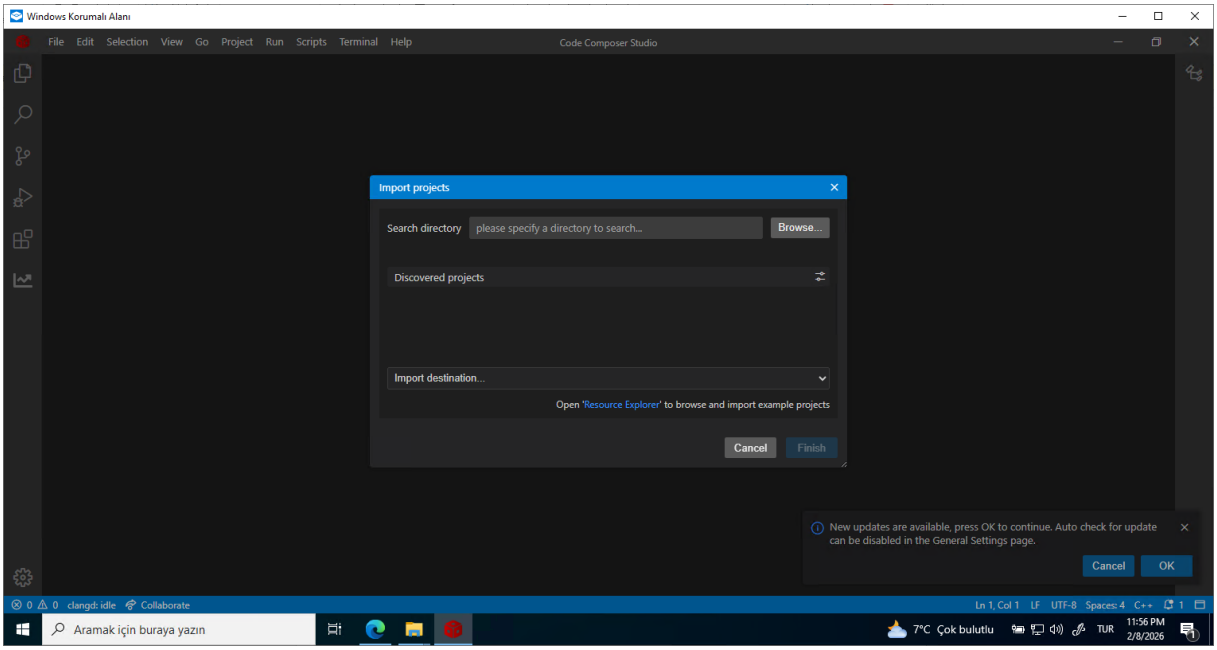


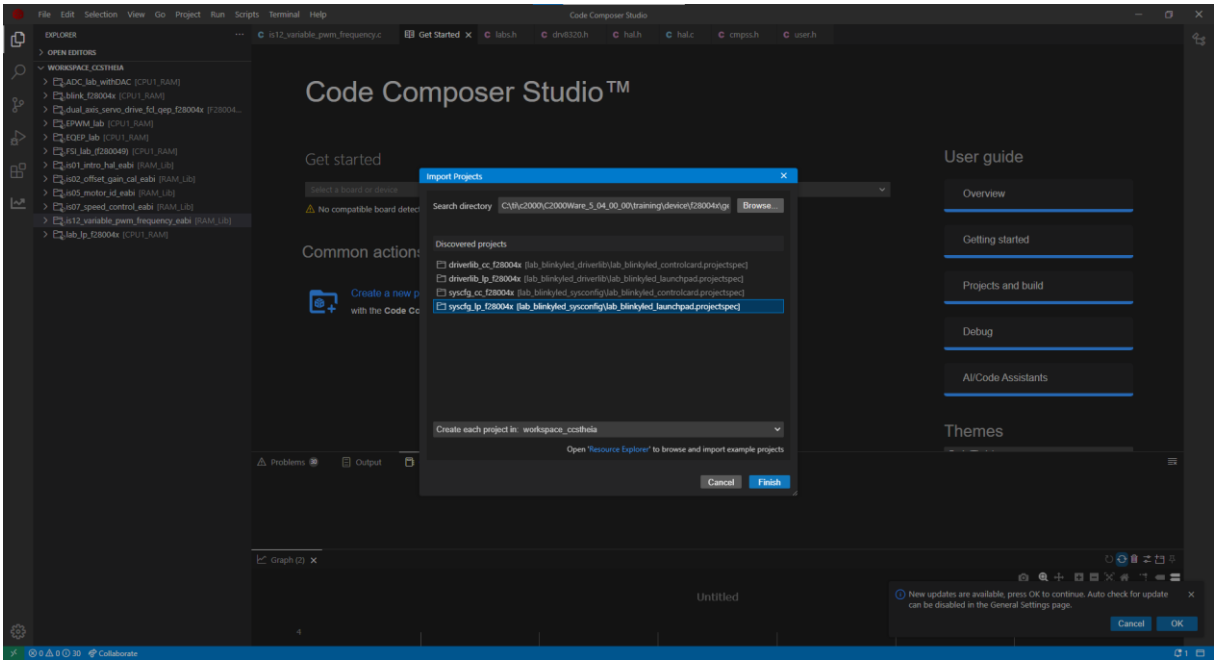
Import a new project from "Import Project" button.



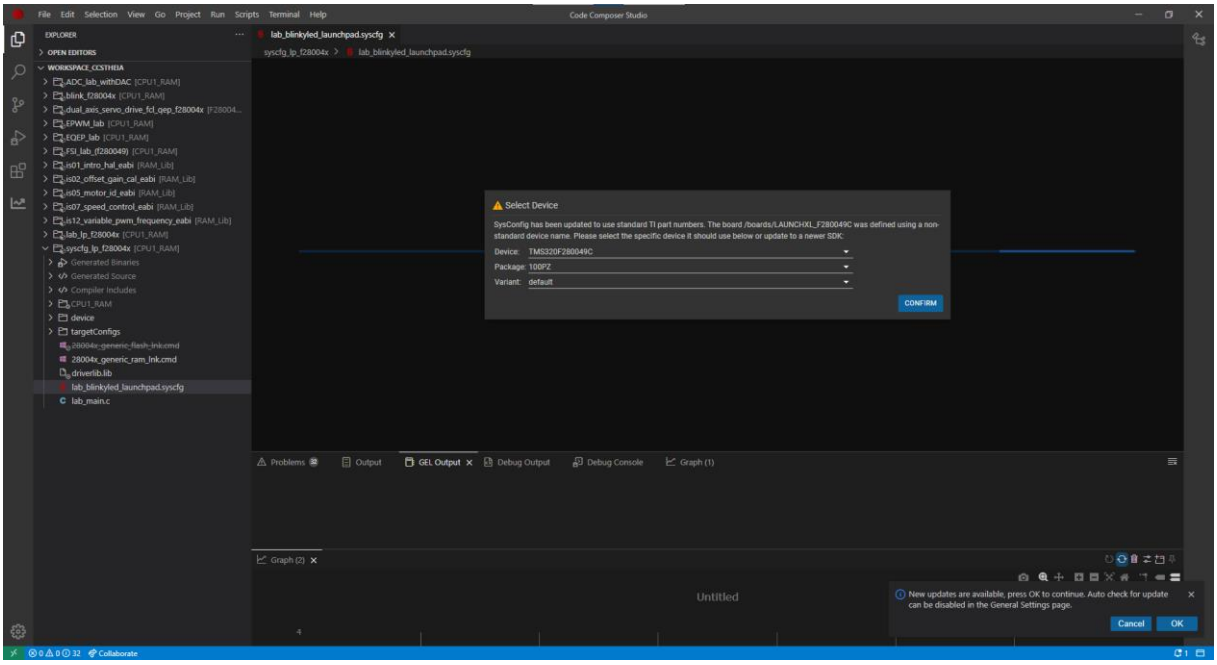
Click "Browse" button and go to " C:\ti\c2000\C2000Ware_5_04_00_00\training\device\f28004x\getting_started" directory and select " syscfg_lp_f28004x". (This is for my case since our custom board uses TMS320F280049 device. Select your own device folder from

"C:\ti\c2000\C2000Ware_5_04_00_00\training\device\" and select getting started folder inside.)

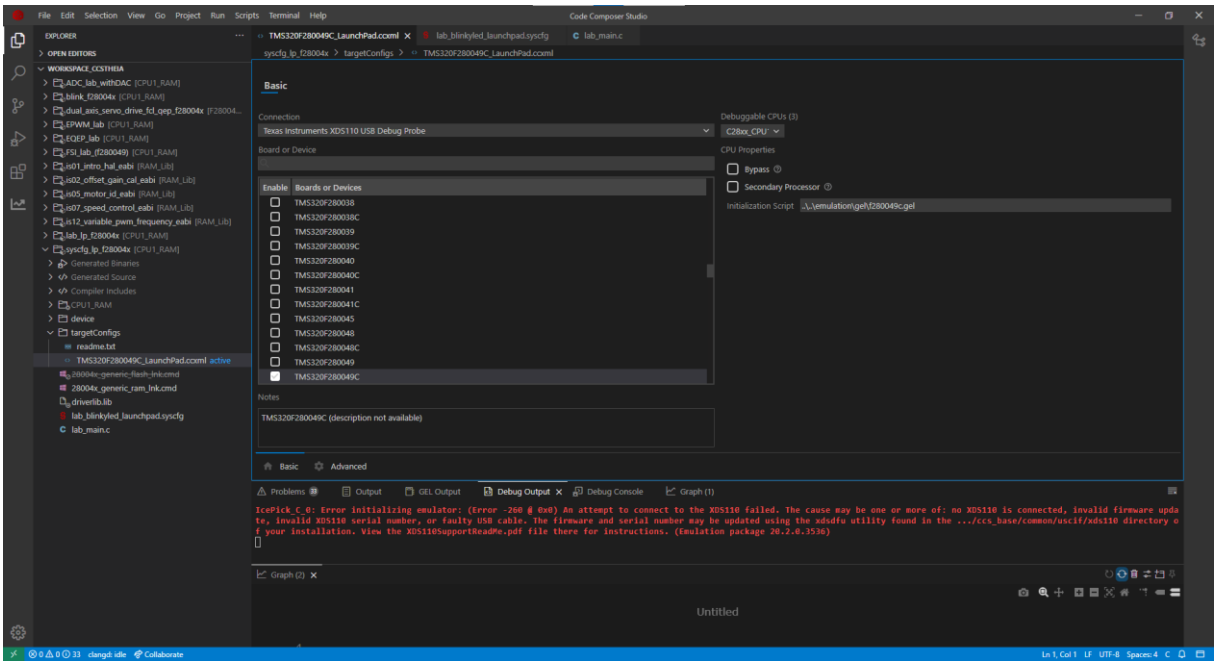




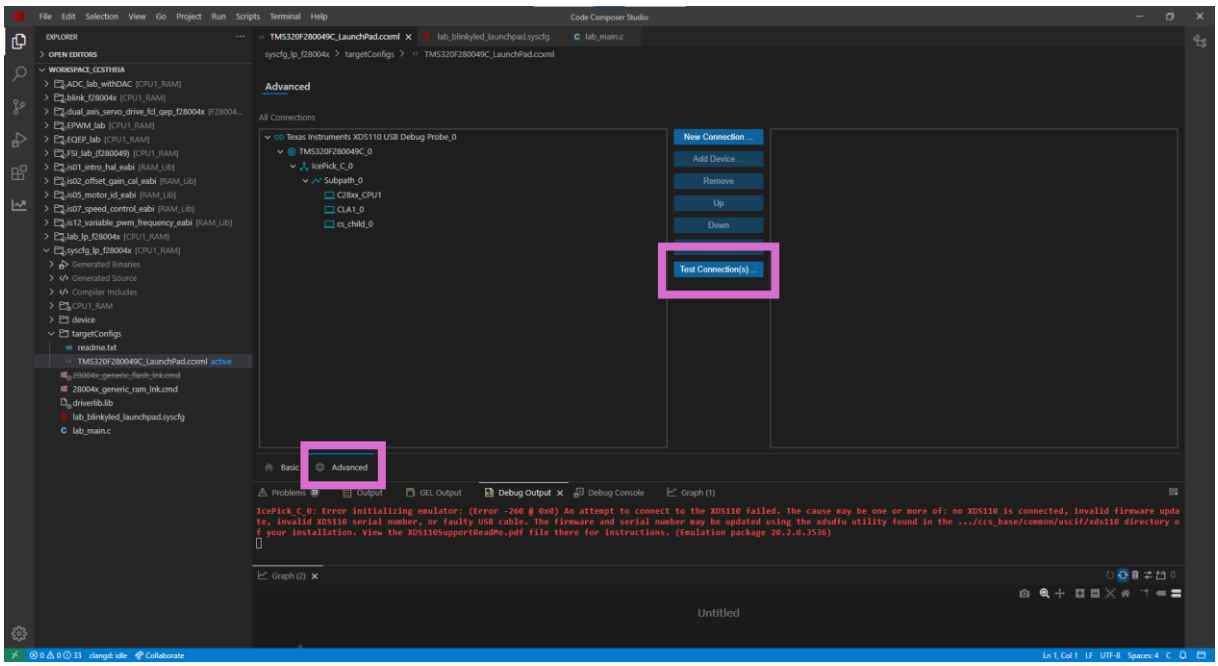
After the project is imported, double click the "lab_blinkyled_launchpad.syscfg" then select your device from the popup menu.



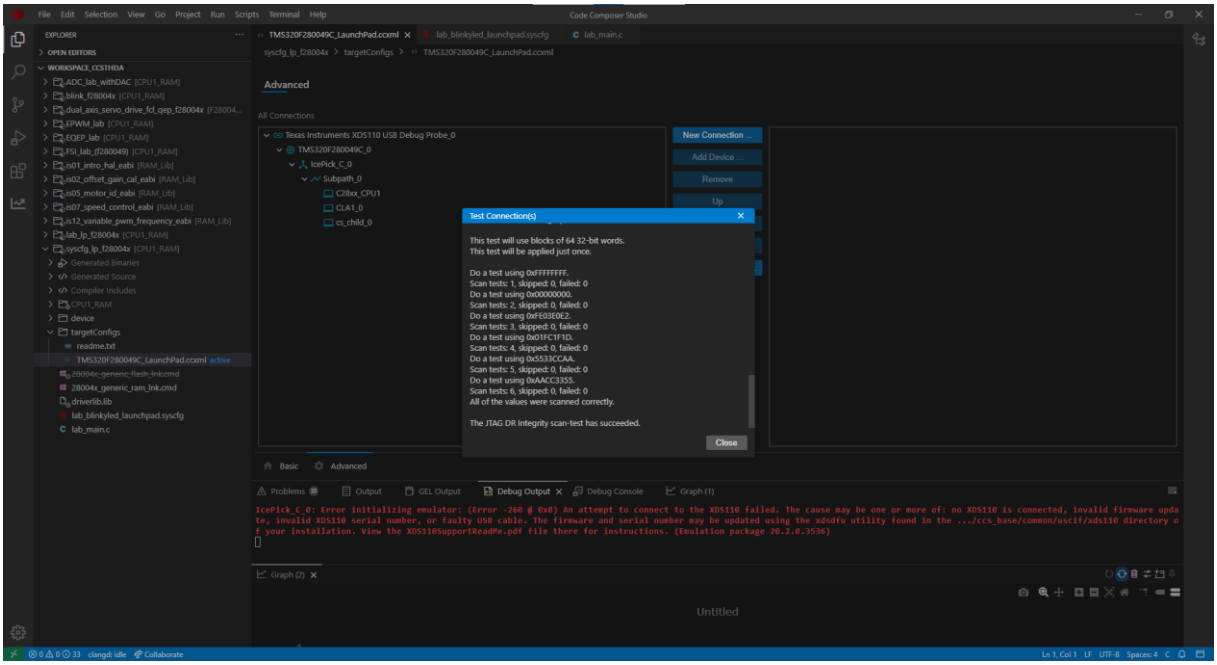
Then to check the JTAG communication double click " TMS320F280049C_LaunchPad.ccxml "



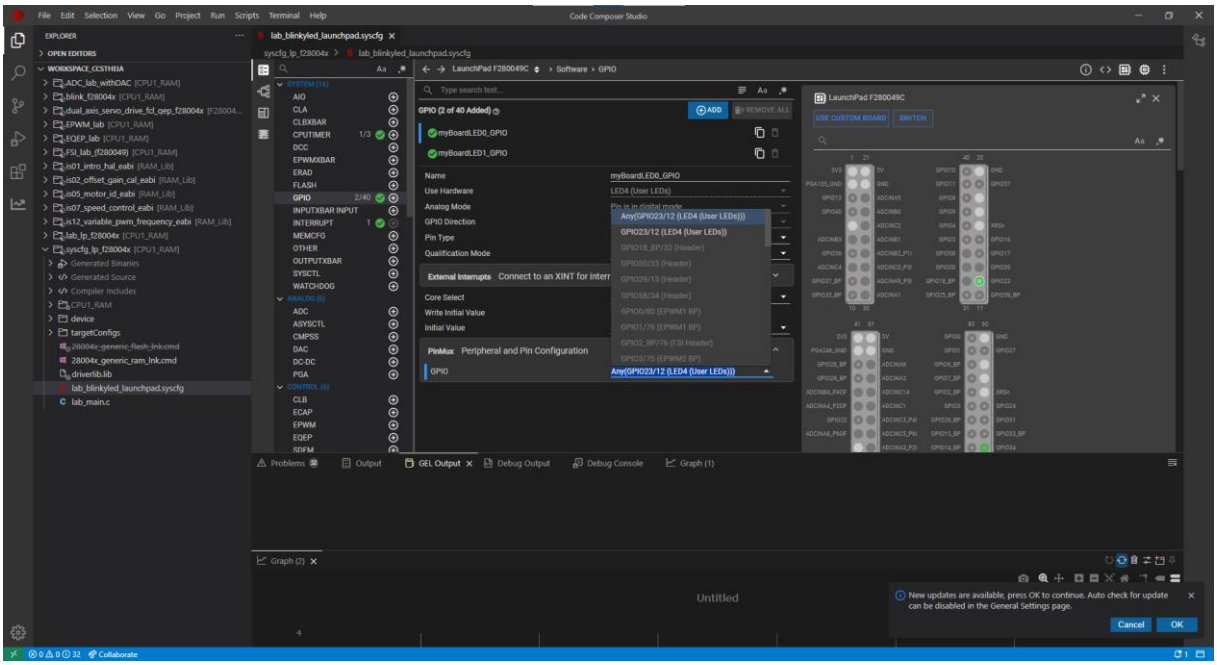
Then click advanced and test connection buttons.



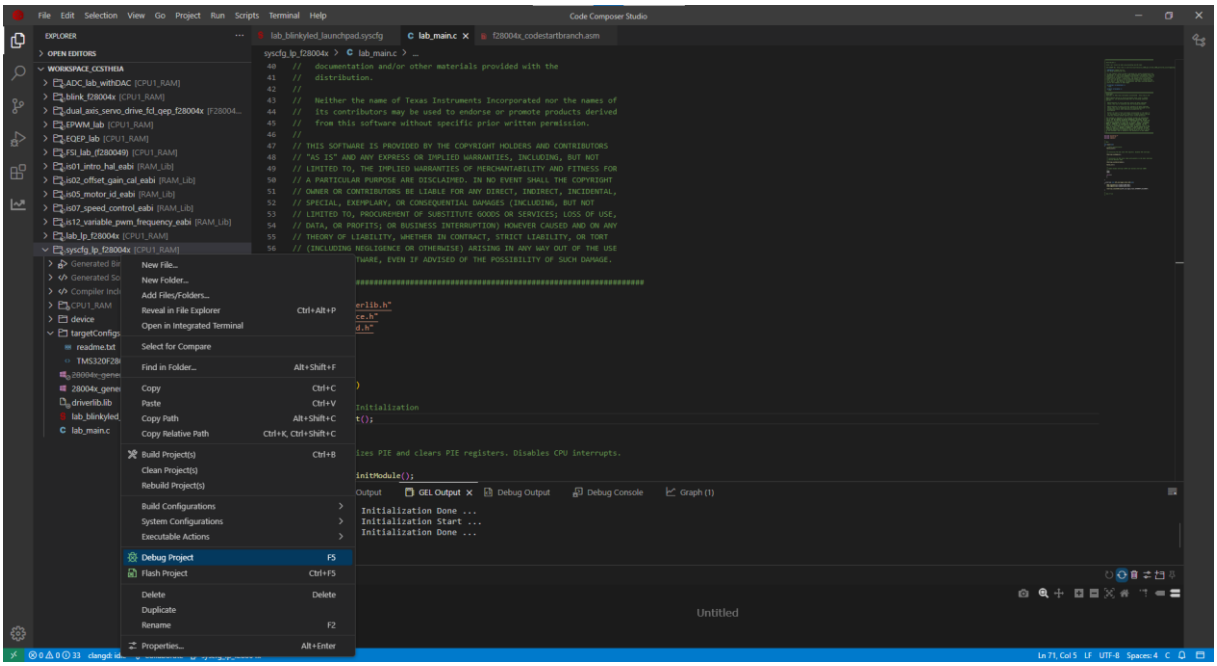
After the test has been completed, it should be seen as follows:



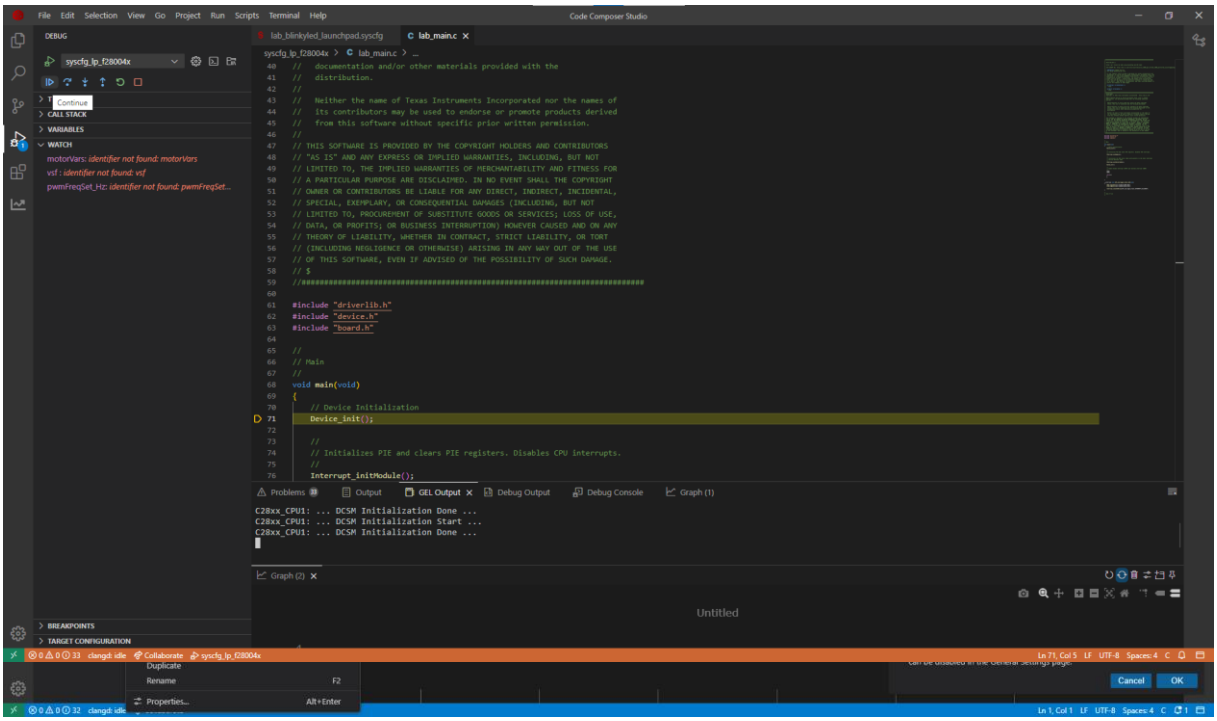
After checking the connection, we can continue with the uploading of the first test code (led blink code). Double click "lab_blinkyled_launchpad.syscfg" then from the GPIO menu, under PinMux select the appropriate GPIO that you connected your led on your own custom board. If you can not find a suitable GPIO from default GPIO's you can add a new GPIO and name it as "myBoardLED0_GPIO" or "myBoardLED1_GPIO". Again under the PinMux select your pin number to assign those pins to your GPIO.



After it is done save sysconfig file and close it. Right click the imported project from file explorer and click debug as shown below:



After the debug session started click continue button as shown below:



After completing those steps you should see that your first code is uploaded and your led should be blinking.